Critical Care Nursing

Final Examination

Part I: Multiple Choice Questions (50 Items)

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1. What is the main function of the buffer system in the body?
A. Increase blood pressure
B. Neutralize bacteria
C. Regulate acid-base balance
D. Produce energy

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2. What is the effect of increased exhalation of CO_2 during metabolic acidosis?
A. Increases the acid level
B. Decreases the acid level
C. Increases blood glucose
D. Has no effect

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3. Which organ compensates more quickly for acid-base imbalance?
A. Kidneys
B. Liver
C. Lungs
D. Pancreas

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4. A patient has the following arterial blood gases: $PaCO_2$ 33, HCO_3 15, pH 7.23. Which condition below is presenting?
A. Metabolic alkalosis partially compensated
B. Metabolic acidosis partially compensated
C. Respiratory alkalosis not compensated
D. Metabolic acidosis fully compensated

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5. Which of the following patients should be given the highest priority during triage?
A. Mild abdominal pain

B. A sprained ankle
C. Chest pain and difficulty breathing
D. A patient with a cold

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6. What is the triage category for a patient who needs immediate life-saving intervention?
A. Non-urgent
B. Urgent
C. Emergent
D. Delayed

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7. Which healthcare professional typically performs triage in the ED?
A. Pharmacist
B. Lab Technician
C. Registered Nurse
D. Radiologist

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8. How does ARDS affect the lungs?
A. It increases the lungs' ability to exchange gases
B. It prevents proper oxygen and carbon dioxide transfer
C. It causes high blood pressure
D. It leads to swelling in the legs

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9. What is the main cause of Acute Respiratory Distress Syndrome (ARDS)?
A. Bacterial infection in the brain
B. Severe fluid buildup in the lungs
C. Low blood sugar
D. Joint inflammation

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10. What clinical consequence results from impaired gas exchange in ARDS?
A. Hyperglycemia
B. Hypoxia

C. Bradycardia
D. Hypercalcemia

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11. Why should nurses avoid overexerting ARDS patients during care?
A. It increases risk of bleeding
B. It can cause dehydration
C. It increases oxygen demand and fatigue
D. It raises blood sugar levels

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12. Which condition is most likely to cause a fat embolism?
A. Amniotic fluid aspiration
B. IV drug use
C. Long bone fracture
D. Central line insertion

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13. Which of the following medications is used to dissolve clots in PE?
A. Bronchodilators
B. Fibrinolytic therapy
C. Sedatives
D. Analgesics

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14. What is the most common source of thrombotic emboli in pulmonary embolism?
A. Pulmonary veins
B. Deep veins of the legs
C. Left atrium
D. Coronary arteries

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15. SIMV is often used for:
A. Initial full ventilatory support
B. Weaning patients from mechanical ventilation
C. Treating respiratory arrest
D. Sedated, paralyzed patients

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16. Which of the following is a common cause of a low-pressure alarm?
A. Patient coughing
B. Circuit or tubing disconnection
C. Bronchospasm
D. Mucus plugging

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17. What is the primary function of Positive End-Expiratory Pressure (PEEP)?
A. To increase the respiratory rate
B. To maintain alveolar recruitment at the end of expiration
C. To cool the air delivered to the lungs
D. To decrease the oxygen concentration

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18. To prevent ventilator-associated pneumonia and aspiration, the head of the patient’s bed should be elevated to:
A. 5–10 degrees
B. 15–20 degrees
C. 30–45 degrees
D. 90 degrees

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19. Shock is best defined as:
A. Decrease in blood pressure
B. Inadequate tissue perfusion leading to cellular dysfunction
C. Failure of the heart to pump blood
D. Decrease in blood volume

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20. The hallmark of all types of shock is:
A. Hypotension
B. Ineffective tissue perfusion
C. Tachycardia
D. Metabolic acidosis

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21. Septic shock is a type of:
A. Cardiogenic shock

B. Hypovolemic shock
C. Distributive shock
D. Obstructive shock

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22. In the initial stage of shock, cells switch from:
A. Aerobic to anaerobic metabolism
B. Anaerobic to aerobic metabolism
C. Protein to fat metabolism
D. Glucose to lipid metabolism

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23. Fluid resuscitation in hypovolemic shock is best initiated with:
A. Blood transfusion
B. Colloid solution
C. Crystalloid solution (e.g., normal saline)
D. Dextrose solution

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24. Paradoxical chest wall motion is:
A. Outward movement during inspiration
B. Inward movement during inspiration
C. Normal chest wall movement
D. Chest expansion is equal

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25. Flail chest is defined as:
A. One rib fracture
B. Three or more ribs fractured in two or more places
C. Any rib fracture with lung contusion
D. Rib fracture with hemothorax

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26. The hallmark sign of tension pneumothorax is:
A. Bradycardia
B. Mediastinal shift
C. Hemoptysis
D. Cough

27. Beck's triad includes:

A. Hypertension, bradycardia, muffled heart sounds

B. Hypotension, jugular vein distension, muffled heart sounds

C. Dyspnea, chest pain, cough

D. Fever, hypotension, tachycardia

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28. The first step in one-person CPR is to:

A. Start chest compressions

B. Check responsiveness and call for help

C. Give 2 breaths

D. Attach AED

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29. What is the correct compression rate for adult CPR?

A. 50–60/min

B. 60–80/min

C. 100–120/min

D. 140–160/min

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30. Which medication is indicated for pulseless ventricular tachycardia or ventricular fibrillation unresponsive to defibrillation?

A. Atropine

B. Epinephrine

C. Amiodarone

D. Lidocaine

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31. Therapeutic hypothermia after cardiac arrest is used to:

A. Stop heart function temporarily

B. Reduce neurological injury

C. Increase blood pressure

D. Replace CPR

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32. Type I respiratory failure is characterized by:

A. Low PaO_2 and high $PaCO_2$

B. Low PaO_2 and normal $PaCO_2$

C. High PaO_2 and low $PaCO_2$

D. High PaO_2 and high $PaCO_2$

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33. Which diagnostic test is gold standard for confirming respiratory failure?

A. Chest X-ray

B. Arterial Blood Gas (ABG)

C. Pulmonary function test

D. Bronchoscopy

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34. Oxygen therapy should maintain SpO_2 at:

A. 70–80%

B. 88–95%

C. 60–75%

D. 100% at all times

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35. Patient positioning in unilateral lung disease should be:

A. Healthy lung in dependent position

B. Diseased lung in dependent position

C. Supine at all times

D. Prone only

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36. How often should incentive spirometry be performed?

A. Once a day

B. At least 10 deep breaths per hour

C. Only when patient is hypoxic

D. Every 4–6 hours

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37. Behavior Pain Scale (BPS) assess which of the following:

a. Facial Expression

b. Upper Limbs

c. Compliance with Ventilation
d. All of the above

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38. Types of pain affect nerve ending of the organs include which of the following:
a. Acute pain
b. Chronic pain
c. Neuropathic pain
d. Nociceptive pain

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39. Chronic pain characterized by:
a. Duration less than 6 months
b. Occur with acute onset
c. Duration more than 6 months
d. Need rapid management

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40. T wave appear in ECG in which phase:
a. Atrial depolarization
b. Ventricular depolarization
c. Atrial repolarization
d. Ventricular repolarization

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41. Normal sinus rhythm characterized by:
a. Present of atrial and ventricular depolarization and repolarization
b. Present of atrial depolarization only
c. Present of ventricular depolarization
d. Present of atrial depolarization and ventricular repolarization

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42. Atrial depolarization indicates which of the following on an electrocardiogram rhythm strip?
a. U waves
b. QRS complex
c. T waves
d. P waves

43. Sinus tachycardia characterized by which of the following:
a. Rapid sinus rhythm
b. Rapid irregular rhythm
c. Slow regular rhythm
d. Regular sinus rhythm

44. Atrial fibrillation characterized by which of the following:
a. Rapid sinus rhythm
b. Slow irregular rhythm
c. Rapid irregular rhythm with absence of P wave
d. Slow regular rhythm

45. Management of ventricular fibrillation (VF) includes:
a. Cardioversion
b. Defibrillation
c. Rate control medication
d. Anticoagulant medication

46. Signs and symptoms of myocardial infarction (MI) include which of the following:
a. Moderate chest pain relieved by sedation
b. Severe chest pain not relieved by sedation
c. Abdominal pain
d. Fever

47. Complications of myocardial infarction (MI) include which of the following:
a. Heart block
b. Ventricular tachycardia
c. Ventricular aneurysm
d. Hypertension

48. Acute treatment of myocardial infarction (MI) include which of the following:
a. Cardioversion

b. Defibrillation
c. Thrombolytic therapy
d. Adenosine

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49. Complication of cardiac surgery:
a. Hypertension
b. Aneurysm
c. Cardiac tamponade
d. Pneumothorax

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50. Indication of cardiac surgery include:
a. Multivessels coronary artery disease
b. Unstable angina
c. Heart block
d. Hypertension

Part II: Short Answer Questions (12 Items)

List the steps in **Arterial Blood Gas (ABG) Interpretation..**

Describe the **Triage priority levels (methods)**.

Outline the **nursing management for a patient with ARDS**.

. Identify the common **causes of Venous Stasis**.

. List the **types of shock** and their respective **causes**.

List common **clinical manifestations of rib fractures**.

List at least **five causes of cardiopulmonary arrest**.

State the **clinical manifestations of Respiratory Failure**.

List **three non-pharmacological treatments of pain**.

Describe the **characteristics of ventricular tachycardia**.

Outline the **medical management of patients with MI**.

Discuss the **preoperative preparation of patients** for **cardiac surgery**.